Mohamed Dhibi

Computer Vision Engineer

Freiburg im Breisgau, Germany

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Summary

Computer Vision and AI Engineer with a strong software engineering background, specializing in real-time optimization and industrial automation. Experienced in bridging specialized camera hardware with deep learning inference to build low-latency, production-ready solutions. Skilled in complex data pipelines, including hyperspectral imaging, backend systems, and DevOps workflows.

Experience

Polysecure GmbH — **Freiburg, Germany**

Computer Vision and Algorithms Engineer — *Dec 2024 – Present*

- Developed VIS/NIR acquisition modules for hyperspectral line-scan polymer sorting, including camera configuration, adapter structure, preprocessing, and integration with classification logic.
- Deployed CNN-based polymer identification and classification workflows, connecting hyperspectral acquisition, preprocessing, model inference, and production decision logic.
- Optimized processing speed for real-time operation, reducing per-item processing time from ∼5 s to ∼0.3 s for a 2 m-long item using compiled C-based functions and acceleration tools such as Numba.
- Implemented color-metric conversion pipelines from hyperspectral data to XYZ and LAB color spaces, enabling RAL-based shade matching and color-distance evaluation.
- Developed frontend–backend communication features with Django REST API endpoints and plain JavaScript, enabling interactive controls and reliable production data exchange.

Pforzheim University — **Pforzheim, Germany**

Data Scientist — *May 2024 – Nov 2024*

- Built a PMMA classifier achieving up to 88% accuracy by combining image-based features with OCR-based text extraction on approximately 1,200 real-world samples from three partner companies in the PACE project.
- Reduced per-sheet inspection time from ∼6 s to ∼1 s by automating evaluation steps that were previously handled manually.
- Deployed the prototype to a server and demonstrated it live, enabling project partners to upload and analyze production data through a central interface.
- Worked closely with academic and industrial stakeholders to adapt the prototype to practical data quality, labeling, and usability requirements.

Biware — **Tunis, Tunisia**

Data Science Intern — *Jul 2023 – Sep 2023*

- Automated data collection from five external websites, scraping 3,000+ articles and more than 10k data points, replacing several days of manual work per reporting cycle.
- Delivered dashboards used by three departments, centralizing KPIs and alerts for recurring reviews.
- Cleaned, structured, and prepared collected data for reporting and business intelligence use cases.

Gafsa Phosphate Company – CPG — **Gafsa, Tunisia**

Computer Technician Intern — *Jun 2022 – Aug 2022*

- Supported maintenance and troubleshooting of up to 10 production machines, helping reduce downtime caused by recurring technical issues.
- Assisted with hardware, software, and network-related support tasks in an industrial environment.

Key Projects

Graph-Based Recommendation Platform for Risk Management *Oct 2023 – Feb 2024*

- Designed and implemented an end-to-end knowledge graph pipeline to convert unstructured risk reports into structured entity-relation networks.
- Developed a **Graph Neural Network (GNN)** to generate context-aware risk recommendations, integrated into a **Django**-based web platform.

CI/CD DevOps Pipeline for Microservices *Sep 2023 – Jan 2024*

- Architected and deployed a production-ready CI/CD pipeline to automate the build, testing, and deployment lifecycle of **Spring Boot** microservices.
- Integrated **Jenkins** for build automation, managed dependencies with **Maven**, and implemented observability using **Grafana** and **Prometheus**.

AI-Driven Job Matching Platform (Jobit) *Jan 2023 – Jun 2023*

- Built a full-stack web platform to aggregate job opportunities from online job boards and offline sources such as emails and local documents.
- Implemented semantic matching algorithms using **Keras/TensorFlow** to match user profiles with relevant job postings through a **Django** backend.

Education

ESPRIT – Private Higher School of Engineering and Technology **Tunis, Tunisia**
M.Sc. in Computer Science – Data Science *Sep 2021 – Nov 2024*
Master's thesis: 18.8/20, Excellent (approx. German grade: 1.3)

Pforzheim University **Pforzheim, Germany**
Exchange Student *Apr 2024 – Oct 2024*

IPEIEM – El Manar Preparatory Engineering Institute **Tunis, Tunisia**
Intensive Engineering Preparatory Program *Sep 2019 – Jul 2021*
CPGE–PTSI: Physics, Technology and Engineering Sciences

Technical Skills

Core Languages: Python, C/C++ (Low-Latency), SQL
Other Languages: Java, C#, JavaScript
AI & Deep Learning: PyTorch, TensorFlow, CNNs, Transformers, Anomaly Detection, Scikit-learn
Computer Vision: OpenCV, Image Processing, Line-Scan, OCR, Color Spaces
Industrial Imaging: VIS/NIR/SWIR Systems, Hyperspectral Data, Acquisition Pipelines
Backend & Data: Django, Flask, Spring Boot, .NET, REST APIs, Spark, NoSQL
Infra & DevOps: Docker, Git, CI/CD, Azure, Grafana, Prometheus, Linux
Languages: English (C2), French (C1), Arabic (C2), German (B1, actively pursuing B2)

Certifications

Nov 2023: Applications of AI for Anomaly Detection – Nvidia
Oct 2023: Building Transformer-Based NLP Applications – Nvidia
Mar 2023: Neural Networks and Deep Learning – DeepLearning.AI